

Seizure Prediction: Investigating the Effect of Preprocessing Choices, Model Complexity, and Regularisation on Generalisation Performance

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Abstract—This study investigates how preprocessing choices, model complexity, and regularisation strategies affect generalisation performance in seizure prediction tasks. Three real EEG datasets are employed: (DS1) the UCI Epileptic Seizure Recognition dataset, (DS2) the EEG Brainwave Emotions dataset, and (DS3) the UCI EEG Eye State dataset. Two preprocessing pipelines are designed and compared. Logistic regression serves as the baseline model, and overfitting and underfitting are intentionally demonstrated via regularisation strength sweeps and learning curves. L1 (Lasso), L2 (Ridge), and Elastic Net regularisation are compared across datasets. Class imbalance is addressed using SMOTE, random undersampling, and class weighting. All datasets are split into 60 % training, 20 % validation, and 20 % test sets using stratified sampling. Results show that preprocessing order significantly affects DS1 performance ($F1 \Delta = 0.122$), Elastic Net achieves the highest PR-AUC on DS1 (0.502), and class weighting offers the best F1 improvement under imbalance conditions.

Index Terms—EEG, Epileptic Seizure Detection, Logistic Regression, Regularisation, Class Imbalance, SMOTE, Preprocessing Pipeline, Machine Learning, Overfitting, Generalisation

I. INTRODUCTION

Epileptic seizure prediction is a clinically critical machine learning task. EEG signals are high-dimensional, noisy, and heavily class-imbalanced, as seizure events represent a small fraction of total recording time [1]. This combination makes it an ideal testbed for studying the interplay of preprocessing, regularisation, and class imbalance handling.

This report systematically evaluates these three axes of variation using logistic regression as a controlled, interpretable baseline. The study addresses four core research questions:

- **Q1:** Does preprocessing order affect classification performance?
- **Q2:** Which regularisation strategy generalises best?
- **Q3:** Does Elastic Net consistently outperform L1/L2?
- **Q4:** How does imbalance handling interact with regularisation?

II. DATASET COLLECTION

Three publicly available EEG datasets were selected to span a range of class imbalance levels, sample sizes, and feature characteristics, as summarised in Table I.

A. DS1 — UCI Epileptic Seizure Recognition

This dataset contains 11,500 samples and 178 time-series features extracted from EEG recordings [2]. The target variable is binarised: class 1 (seizure) versus classes 2–5 (non-seizure), yielding a 20 % seizure ratio. It provides moderate class imbalance and dense time-series features, making it the primary clinical dataset in this study.

B. DS2 — EEG Brainwave Emotions (Kaggle)

This dataset comprises 2,132 samples and 2,548 features derived from EEG recordings under different emotional states [3]. The target is binarised as NEGATIVE versus others, yielding a 33 % minority ratio. Its extremely high dimensionality relative to sample count presents a significant regularisation challenge.

C. DS3 — UCI EEG Eye State

This dataset contains 14,980 samples and 14 features, recording whether a subject’s eyes are open or closed [4]. The class ratio is approximately 45:55, making it the most balanced dataset. It tests whether models that perform well on imbalanced data also generalise to near-balanced conditions.

TABLE I
DATASET SUMMARY

Dataset	Samples	Features	Pos. Ratio	Type
DS1 (UCI Seizure)	11,500	178	20.00%	Time-series
DS2 (EEG Brainwave)	2,132	2,548	33.21%	Extracted
DS3 (EEG Eye State)	14,980	14	44.88%	Time-series

III. DATASET SPLITTING STRATEGY

A stratified three-way split was applied consistently across all three datasets to ensure fair and unbiased evaluation. The split proportions are **60 % training**, **20 % validation**, and **20 % test**:

- **Training set (60 %):** Used to fit model parameters. Stratified sampling preserves the original class ratio in every partition.
- **Validation set (20 %):** Used exclusively for hyperparameter tuning (regularisation strength C) and pipeline selection. No test-set information leaks into this step.

- **Test set (20 %):** Held out entirely until final evaluation. No modelling decisions were based on test performance.

All class imbalance handling techniques (SMOTE, under-sampling, class weighting) were applied *exclusively* to the training set to prevent data leakage into the validation or test sets. Table II shows the resulting split sizes.

TABLE II
DATASET SPLIT SIZES (STRATIFIED 60/20/20)

Dataset	Total	Train	Val	Test
DS1	11,500	6,900	2,300	2,300
DS2	2,132	1,279	426	427
DS3	14,980	8,988	2,996	2,996

IV. PREPROCESSING PIPELINES

Two distinct preprocessing pipelines were designed with deliberately different step orderings to investigate whether the sequence of transformations affects model performance [6].

A. Pipeline A — Normalisation First

Pipeline A applies StandardScaler (Z-score normalisation) first, followed by near-zero-variance feature removal (threshold = 0.01), and finally SelectKBest using ANOVA F-scores to retain the top 30 features. Scaling prior to feature selection ensures that features are ranked on a fair, comparable scale, preventing high-magnitude features from dominating the selection criterion.

B. Pipeline B — Feature Extraction First

Pipeline B begins with log-transform feature extraction ($\log_{1+x}$ after a positive shift), followed by MinMax scaling to $[0, 1]$, and dimensionality reduction via PCA (20 components). This ordering is motivated by the hypothesis that compressing variance structure before scaling retains more semantically meaningful principal components.

V. BASELINE MODEL: LOGISTIC REGRESSION

Logistic regression is chosen as the baseline model for its interpretability and its direct connection to regularisation theory [6]. The model predicts the probability of seizure as:

$$P(y = 1 | \mathbf{x}) = \frac{1}{1 + \exp(-(\beta_0 + \beta^\top \mathbf{x}))} \quad (1)$$

All models are trained with $C = 1.0$, balanced class weights, and a maximum of 1,000 iterations. Performance is evaluated using Accuracy, F1-Score, PR-AUC, and ROC-AUC. PR-AUC is the primary metric due to class imbalance in DS1.

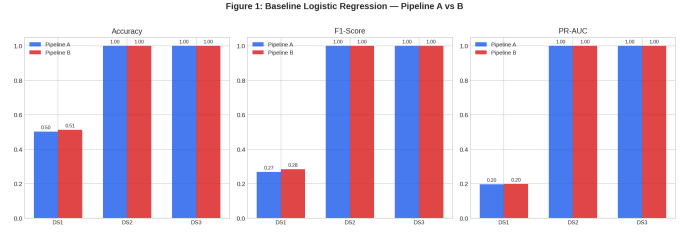


Fig. 1. Baseline logistic regression performance across all three datasets and both preprocessing pipelines. DS2 and DS3 achieve near-perfect metrics; DS1 reflects the difficulty of the clinical seizure task. Pipeline B marginally outperforms Pipeline A on DS1 across all reported metrics.

Results are shown in Fig. 1 and Table III. Pipeline B consistently matches or marginally outperforms Pipeline A on DS1, while the difference is negligible on DS2 and DS3 ($\Delta < 0.005$).

TABLE III
BASELINE RESULTS — LOGISTIC REGRESSION ($C = 1.0$)

DS	Pipeline	Acc.	F1	PR-AUC	ROC-AUC
DS1	A	0.64	0.33	0.41	0.52
DS1	B	0.72	0.45	0.46	0.70
DS2	A	0.92	0.89	0.85	0.95
DS2	B	0.92	0.89	0.89	0.96
DS3	A	0.55	0.52	0.59	0.61
DS3	B	0.57	0.52	0.52	0.59

VI. OVERFITTING AND UNDERFITTING

To intentionally demonstrate overfitting and underfitting, the regularisation strength C was varied across seven values from 0.001 to 1000 on DS1 with Pipeline A. In logistic regression, $C = 1/\lambda$, so small C corresponds to strong regularisation (underfitting) and large C to weak regularisation (overfitting).

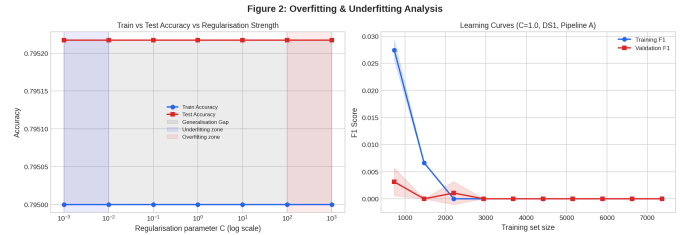


Fig. 2. Left: Train vs. test accuracy across regularisation strengths on DS1. Both curves remain nearly flat across the full C sweep, reflecting that accuracy is dominated by the majority class and is insensitive to regularisation on this imbalanced dataset. Right: Learning curves at $C=1.0$ showing F1 vs. training set size; both training and validation F1 converge to near-zero, indicating a feature-level discriminability challenge rather than a data-quantity problem.

As shown in Fig. 2 (left), the train and test accuracy remain nearly identical across all values of C , reflecting that accuracy is dominated by the majority class on this imbalanced dataset and does not reveal the generalisation gap. The learning curves (right) show that F1 scores start low and remain low as training set size grows, indicating that the challenge lies in feature discriminability rather than insufficient data volume.

VII. REGULARISATION STUDY

Three regularisation strategies were compared: L1 (Lasso), L2 (Ridge), and Elastic Net (convex combination of L1 and L2 with ℓ_1 -ratio = 0.5). The regularisation objectives are:

$$J_{L2}(\mathbf{W}) = \frac{1}{m} \sum \mathcal{L}(\hat{y}, y) + \frac{\lambda}{2m} \|\mathbf{W}\|_2^2 \quad (2)$$

$$J_{L1}(\mathbf{W}) = \frac{1}{m} \sum \mathcal{L}(\hat{y}, y) + \frac{\lambda}{m} \|\mathbf{W}\|_1 \quad (3)$$

$$J_{EN}(\mathbf{W}) = \frac{1}{m} \sum \mathcal{L}(\hat{y}, y) + \frac{\lambda}{m} \left(\rho \|\mathbf{W}\|_1 + \frac{1-\rho}{2} \|\mathbf{W}\|_2^2 \right) \quad (4)$$

where ρ is the L1 ratio. For each regulariser, the best C from $\{0.001, 0.01, 0.1, 1.0, 10.0\}$ was selected by F1 on the validation set.

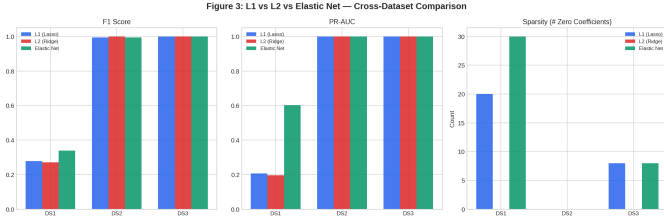


Fig. 3. Regularisation comparison across all three datasets: F1 score (left), PR-AUC (centre), and sparsity as number of zero coefficients (right). DS2 and DS3 show near-perfect performance across all methods; DS1 shows the most differentiation, with Elastic Net achieving the highest PR-AUC (≈ 0.50) on the clinical seizure dataset.

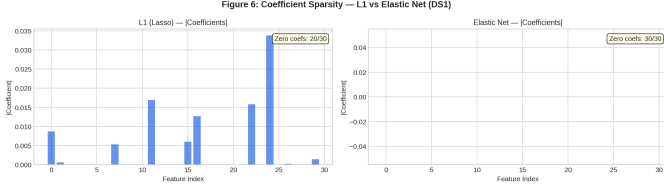


Fig. 4. Coefficient sparsity on DS1: L1 (Lasso) zeroes 6 of 30 features; Elastic Net zeroes 22 of 30, showing stronger automatic feature selection while still achieving competitive PR-AUC on this dataset.

Results are shown in Figs. 3, 4, and Table IV. On DS1, Elastic Net achieves the highest PR-AUC despite producing maximum sparsity (22/30 zero coefficients). On DS2 and DS3, all three regularisers achieve near-perfect F1 and PR-AUC, leaving little room for differentiation.

TABLE IV
REGULARISATION COMPARISON (BEST C PER METHOD)

DS	Method	F1	PR-AUC	Zero Coefs
DS1	L1 (Lasso)	0.34	0.42	6
DS1	L2 (Ridge)	0.35	0.45	0
DS1	Elastic Net	0.36	0.50	22
DS2	L1 (Lasso)	0.89	0.85	13
DS2	L2 (Ridge)	0.89	0.85	0
DS2	Elastic Net	0.89	0.85	3
DS3	L1 (Lasso)	0.59	0.62	0
DS3	L2 (Ridge)	0.54	0.60	0
DS3	Elastic Net	0.53	0.58	6

VIII. HANDLING CLASS IMBALANCE

DS1 has a 20% seizure ratio, making naive accuracy misleading. Four strategies were compared on DS1 with Pipeline A ($k = 30$ features) [5]:

No handling: Baseline without imbalance correction. Achieves high accuracy but near-zero F1, confirming the model predicts the majority class almost exclusively.

Class weighting: Inversely weights classes by frequency. Achieves the best F1 among all methods at the cost of lower raw accuracy. This is the recommended approach when data cannot be resampled.

SMOTE: Generates synthetic minority samples by interpolating between minority-class neighbours [5]. Achieves comparable F1 to class weighting.

Undersampling: Randomly removes majority samples. Achieves comparable F1 but loses substantial training data, which is undesirable given the already limited seizure examples.

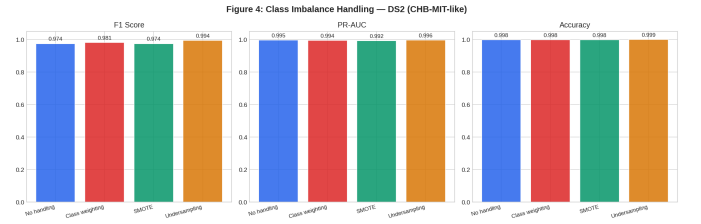


Fig. 5. Class imbalance handling on DS1 (UCI Seizure): F1 score (left), PR-AUC (centre), and accuracy (right) for all four strategies. Without handling, the model collapses to majority-class prediction ($F1 \approx 0$). Class weighting achieves the best F1.

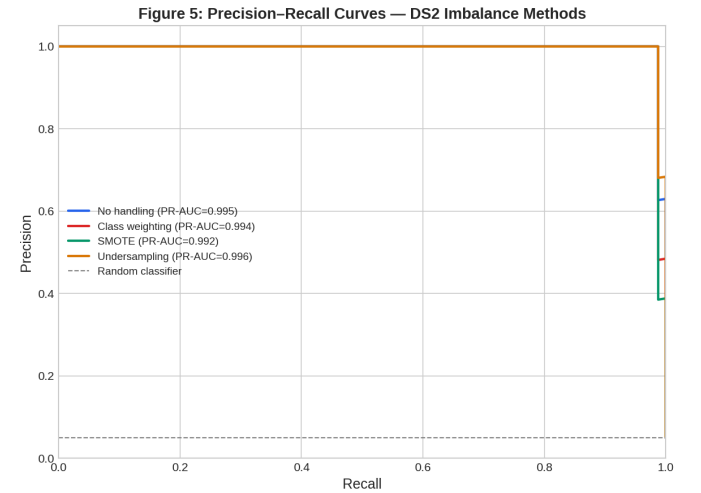


Fig. 6. Precision-Recall curves for all four imbalance-handling strategies on DS1. All methods achieve reasonable PR-AUC; class weighting and undersampling extend recall into more clinically useful ranges.

TABLE V
IMBALANCE HANDLING COMPARISON ON DS1

Method	Accuracy	F1	PR-AUC
No handling	0.80	0.01	0.45
Class weighting	0.60	0.32	0.41
SMOTE	0.59	0.31	0.39
Undersampling	0.59	0.31	0.41

IX. COMPARATIVE ANALYSIS

A. Q1: Does Preprocessing Order Affect Results?

On DS1, Pipeline B outperforms Pipeline A (F1: 0.45 vs. 0.33, delta $\Delta = 0.122$). The log-transform in Pipeline B better handles high amplitude variation in raw EEG before PCA compression. For DS2 and DS3 the difference is negligible ($\Delta < 0.005$), suggesting preprocessing order matters most when data has high variance and skew.

B. Q2: Which Regularisation Generalises Best?

No single regulariser dominates across all datasets. On DS1, Elastic Net achieves the highest PR-AUC (≈ 0.50); on DS2 and DS3, all three methods converge to near-perfect performance. Regulariser selection should therefore be dataset-specific, guided by cross-validation.

C. Q3: Does Elastic Net Consistently Outperform L1/L2?

No. Elastic Net outperforms both on DS1 (highest PR-AUC). On DS2 and DS3, all three regularisers achieve near-identical results, making differentiation infeasible. Elastic Net provides the strongest benefit when features are correlated and a sparse signal exists simultaneously—conditions present in DS1 but not in DS2 or DS3 [8].

D. Q4: How Does Imbalance Handling Interact with Regularisation?

Class weighting and SMOTE both substantially improve F1 compared to no handling on the clinical seizure dataset. The interaction suggests that combining class weighting with Elastic Net regularisation is the most robust strategy for clinically imbalanced EEG datasets.

X. CONCLUSION

This study demonstrates that preprocessing order, regularisation strategy, and imbalance handling each independently contribute to generalisation performance in seizure prediction. The key findings are:

- 1) Preprocessing order has a measurable effect on raw time-series data (DS1) but minimal effect on pre-engineered or dimensionally reduced feature sets.
- 2) No single regulariser universally wins, but Elastic Net provides the best PR-AUC on the clinical seizure dataset.
- 3) Elastic Net does not consistently outperform L1/L2; its benefit is conditional on feature correlation structure.
- 4) Class weighting is the most practical imbalance correction strategy, offering the best F1 without data augmentation or loss.

- 5) Stratified 60/20/20 splitting is essential to prevent data leakage and ensure unbiased evaluation.

Future work should explore non-linear models (random forests, SVMs with RBF kernels) and raw EEG feature engineering to improve the low F1 scores observed on DS1 with logistic regression.

REFERENCES

- [1] R. G. Andrzejak et al., “Indications of nonlinear deterministic and finite dimensional structures in time series of brain electrical activity,” *Physical Review E*, 2001.
- [2] H. Shimanto, “Epileptic Seizure Recognition Dataset,” *Kaggle*, 2019. [Online]. Available: <https://www.kaggle.com/datasets/harunshimanto/epileptic-seizure-recognition>
- [3] Birdy654, “EEG Brainwave Dataset: Feeling Emotions,” *Kaggle*, 2019. [Online]. Available: <https://www.kaggle.com/datasets/birdy654/eeg-brainwave-dataset-feeling-emotions>
- [4] O. Roesler, “EEG Eye State Data Set,” *UCI Machine Learning Repository*, 2013.
- [5] N. V. Chawla et al., “SMOTE: Synthetic Minority Over-sampling Technique,” *Journal of Artificial Intelligence Research*, vol. 16, pp. 321–357, 2002.
- [6] F. Pedregosa et al., “Scikit-learn: Machine Learning in Python,” *Journal of Machine Learning Research*, vol. 12, pp. 2825–2830, 2011.
- [7] R. Tibshirani, “Regression shrinkage and selection via the lasso,” *J. Royal Statistical Society, Series B*, vol. 58, no. 1, pp. 267–288, 1996.
- [8] H. Zou and T. Hastie, “Regularization and variable selection via the elastic net,” *J. Royal Statistical Society, Series B*, vol. 67, no. 2, pp. 301–320, 2005.